

Matrix free Mg^{2+} and the regulation of mitochondrial volume

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Jung, Dennis W., and Gerald P. Brierley. Matrix free Mg^{2+} and the regulation of mitochondrial volume. *Am. J. Physiol.* 277 (Cell Physiol. 46): C1194–C1201, 1999.—Mitochondria must maintain volume homeostasis in order to carry out oxidative phosphorylation. It has been postulated that the concentration of free Mg^{2+} ($[\text{Mg}^{2+}]$) serves as the sensor of matrix volume and regulates a K^+/H^+ antiport (K. D. Garlid. *J. Biol. Chem.* 255: 11273–11279, 1980). To test this hypothesis, the fluorescent probe furaptra was used to monitor $[\text{Mg}^{2+}]$ and free Ca^{2+} concentration ($[\text{Ca}^{2+}]$) in the matrix of isolated beef heart mitochondria, and K^+/H^+ antiport activity was measured by passive swelling in potassium acetate. Concentrations that result in 50% inhibition of maximum activity of 92 μM matrix $[\text{Mg}^{2+}]$ and 2.2 μM $[\text{Ca}^{2+}]$ were determined for the K^+/H^+ antiport. Untreated mitochondria average 670 μM matrix $[\text{Mg}^{2+}]$, a value that would permit <1% of maximum K^+/H^+ antiport activity. Hypotonic swelling results in large decreases in matrix $[\text{Mg}^{2+}]$, but swelling due to accumulation of acetate salts does not alter $[\text{Mg}^{2+}]$. Swelling in phosphate salts decreases matrix $[\text{Mg}^{2+}]$, but not to levels that permit appreciable antiport activity. We conclude that 1) it is unlikely that matrix $[\text{Mg}^{2+}]$ serves as the mitochondrial volume sensor, 2) if K^+/H^+ antiport functions as a volume control transporter, it is probably regulated by factors other than $[\text{Mg}^{2+}]$, and 3) alternative mechanisms for mitochondrial volume control should be considered.

potassium-hydrogen antiport; heart mitochondria; mitochondrial free magnesium concentration; mitochondrial volume control; furaptra

MITOCHONDRIA IN SITU maintain a high negative membrane potential ($\Delta\psi$) in an intracellular milieu containing >140 mM K^+ . Mitchell (35) recognized that the electrophoretic inward movement of K^+ would constitute a threat to the osmotic integrity of the mitochondrion and that a mechanism for volume control was necessary. He proposed that an electroneutral K^+/H^+ antiporter must be present to extrude excess entering K^+ . The presence of such a transporter initially proved difficult to demonstrate, but it is now clear that K^+/H^+ antiport activity can be unmasked when mitochondrial Mg^{2+} is depleted and the pH elevated in hypotonic media (see Refs. 9 and 15 for reviews). In addition, a K^+/H^+ antiporter has been isolated and reconstituted (33), and a brief report of its cloning and partial sequencing has appeared (45).

The permeability of mitochondria to K^+ increases with increasing $\Delta\psi$ in vitro (12), and an ATP-sensitive

electrophoretic K^+ channel is also present (see Ref. 16 for review). It is now thought that electrophoretic K^+ influx and electroneutral efflux must be regulated to maintain mitochondrial volume homeostasis (16). An Mg^{2+} -sensitive electrophoretic anion channel (4) is also present and is believed to participate in net salt extrusion and osmotic contraction.

Garlid (14, 15) has proposed that the mitochondrial K^+/H^+ antiporter is regulated by the matrix free Mg^{2+} concentration ($[\text{Mg}^{2+}]$). In this “ Mg^{2+} carrier-brake” model, matrix $[\text{Mg}^{2+}]$ is postulated to serve as the volume sensor. Mitochondrial swelling due to the uptake of K^+ salts of Mg^{2+} -liganding anions, such as citrate and phosphate, would decrease matrix free $[\text{Mg}^{2+}]$ and promote the disassociation of Mg^{2+} bound to negative regulatory sites on the K^+/H^+ antiport and the anion channel. The resulting activation of these two transporters would bring about net salt efflux and osmotic contraction. This system is seen as adjusting the steady-state rate of K^+/H^+ antiport to equal the rate of electrophoretic K^+ uniport so that matrix volume remains unchanged (15).

Although this elegant model was proposed several years ago, evidence that matrix $[\text{Mg}^{2+}]$ has properties consistent with those of a volume sensor is inconclusive (9, 21, 27), and a precise relationship between matrix $[\text{Mg}^{2+}]$ and K^+/H^+ antiport has yet to be established. It is well known that mitochondria can take up and extrude Mg^{2+} by respiration-dependent processes and that matrix $[\text{Mg}^{2+}]$ can vary with metabolic conditions (see Ref. 27 for review). It is also clear that the antiport is reversibly inhibited by $[\text{Mg}^{2+}]$ (15). However, to confirm that the antiport is regulated by matrix $[\text{Mg}^{2+}]$, it must be shown that matrix $[\text{Mg}^{2+}]$ varies in the concentration range that controls antiport activity (18). To confirm that matrix $[\text{Mg}^{2+}]$ acts as a mitochondrial volume sensor, its concentration should be shown to vary in an appropriate way with volume changes.

The availability of the fluorescent probe furaptra as an indicator of free $[\text{Mg}^{2+}]$ (39) makes a direct test of these issues feasible. This laboratory has defined the properties of furaptra when it is sequestered in the mitochondrial matrix (25, 28) and, more recently, has used this probe to examine the relationship between total mitochondrial Mg^{2+} and matrix $[\text{Mg}^{2+}]$ (29). The present study evaluates matrix $[\text{Mg}^{2+}]$ changes with mitochondrial swelling under three different conditions and compares the rate of K^+/H^+ antiport, as measured by mitochondrial swelling in a hypotonic potassium acetate medium, with matrix $[\text{Mg}^{2+}]$, as reported by the fluorescence of furaptra. The results indicate that matrix $[\text{Mg}^{2+}]$ does not change in a way that is consistent with the role of a volume sensor or that of a

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regulator of K^+/H^+ antiport. It is concluded that alternative mechanisms for sensing mitochondrial volume and regulating osmotic contraction should be considered.

EXPERIMENTAL PROCEDURES

Furaptra loading and divalent cation depletion of mitochondria. Beef heart mitochondria were prepared as previously described (10) and suspended in 0.25 M sucrose-5 mM TES, pH 7.4, at 25 mg of protein per milliliter. Furaptra (also called mag-fura 2) was loaded into the mitochondrial matrix by incubating 25 mg of mitochondria in 2 ml of sucrose-TES containing 0.1 mM EGTA, 20 mM NaCl, 1 mM ATP sodium salt, and 8 μ M furaptra-AM (Molecular Probes) at 24°C. After 18 min, 6 ml of a KCl medium (100 mM KCl, 15 mM HEPES, 1 mM EGTA, 1 mM EDTA, pH 7.4) were added, then 2 μ M BrA-23187 was added to deplete the mitochondria of divalent cations. After an additional 5-min incubation, the mitochondria were diluted with 8 ml of ice-cold KCl medium and collected by centrifugation (11,000 *g* for 10 min). The pellets were resuspended in 10 ml of a KCl medium containing less EGTA (100 μ M) and no EDTA, reisolated (11,000 *g* for 10 min), and resuspended at 25 mg/ml in KCl-HEPES containing 15 μ M EGTA. During the depletion procedure, endogenous Mg^{2+} is reduced from 30–40 nmol/mg protein to ~2 nmol/mg, as estimated by atomic absorption analysis (8). Furaptra loading of mitochondria without cation depletion was carried out as previously described (25, 26). Ionophores were obtained from Calbiochem.

Swelling measurements. Passive osmotic swelling was followed as the light-scattering change at 520 nm with a Brinkman PC801 colorimeter, as previously described (11). The light-scattering variable β was calculated from reciprocal absorbance, as described by Beavis et al. (5). The swelling records shown in Fig. 5A were made using an SLM DW-2C spectrophotometer in the split-beam mode.

Fluorescence measurements. The fluorescence of furaptra was measured as emission at 510 nm with excitation ratios taken at 340/380 nm with a Perkin-Elmer LS-5B fluorometer (25). The fluorescence can be calibrated in terms of matrix $[Mg^{2+}]$ or Ca^{2+} concentration ($[Ca^{2+}]$) by the ratios method (19), since Ca^{2+} binding to furaptra leads to changes in the fluorescence spectra that are quite similar to those due to Mg^{2+} (39). Because the background fluorescence of mitochondria is significant relative to that of furaptra, each experiment was repeated using mitochondria that did not contain furaptra, and net fluorescence intensities (furaptra-loaded – nonloaded) were used to form ratios before calculating $[Mg^{2+}]$ or $[Ca^{2+}]$, as previously described (25, 26).

The dissociation constant (K_d) for Mg^{2+} -furaptra is very dependent on ionic strength. The values determined by Jung et al. (28) of 1.29 mM in 55 mM and 2.1 mM in 100 mM K^+ salts, respectively, were used to calculate matrix $[Mg^{2+}]$. The K_d of Ca^{2+} -furaptra in 55 mM K^+ salts did not appear to be available. A value of 13.7 μ M was determined by recording excitation spectra at pH 7.8 and 25°C in a 55 mM KCl medium containing 5 mM TES, 2 mM nitrilotriacetic acid, 227 nM K^+ -furaptra, and variable $CaCl_2$ (0–2.52 mM) to give $[Ca^{2+}]$ from 0 to 0.6 mM. The $[Ca^{2+}]$ was calculated using the program of Schoenmakers et al. (42) and taking the apparent log stability constant for Ca^{2+} -nitrilotriacetic acid to be 6.80 for the conditions specified.

The basic reaction medium contained 55 mM potassium acetate, 5 mM TES (pH 7.8, 25°C), 30 μ M EGTA, 2 μ g/ml oligomycin, 2 μ g/ml rotenone, and 0.5 mg/ml furaptra-loaded, divalent cation-depleted mitochondria. This corresponds

closely to the 110 mosM medium of Nakashima and Garlid (37). Other additions or modifications are noted in the figure legends.

Matrix pH was determined using the fluorescence of carboxysemaphthorhodofluor-1 (SNARF-1), as described by Baysal et al. (3), or 2',7'-bis(2-carboxyethyl)-5(6)-carboxyfluorescein (BCECF), as described by Brierley et al. (11).

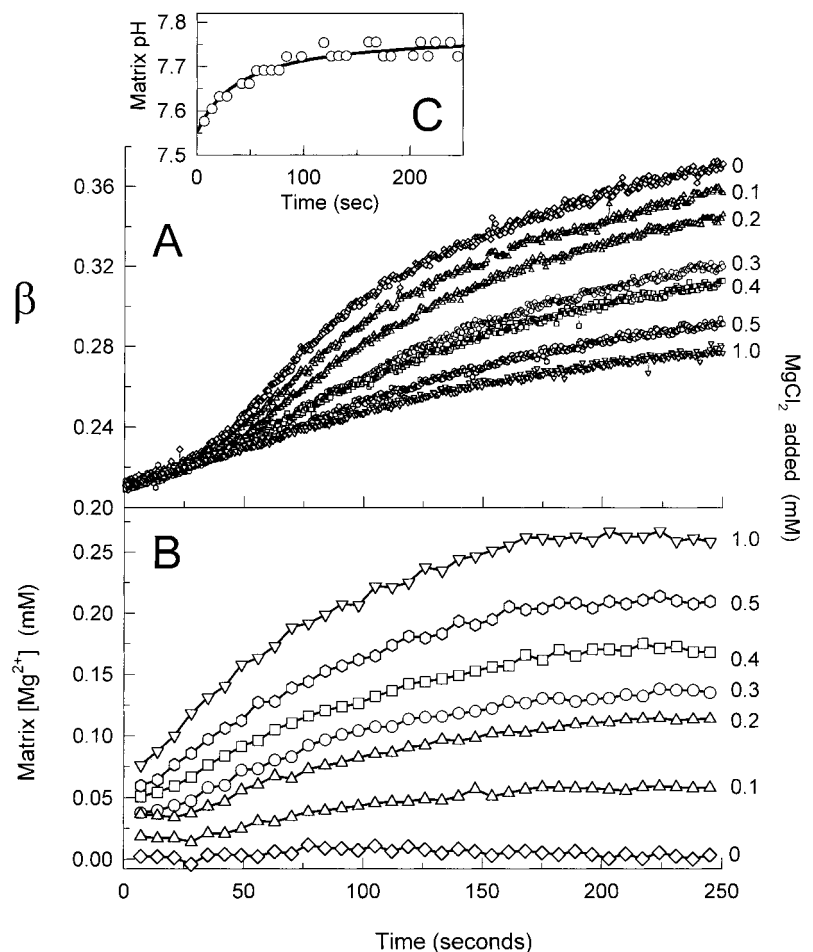
RESULTS

Inhibition of K^+/H^+ antiport by $[Mg^{2+}]$. It has been established that passive swelling in potassium acetate is a convenient and reliable measure of K^+/H^+ antiport activity (17, 37). Swelling is due to the rapid uptake of acetic acid, its ionization, and the exchange of matrix H^+ for external K^+ on the activated K^+/H^+ antiport to give a net accumulation of potassium acetate (17). Beef heart mitochondria depleted of divalent cations swell passively when suspended in this medium (10), which indicates that the K^+/H^+ antiport has been activated. There is essentially no swelling in the absence of divalent cation depletion (10).

The swelling is inhibited in a concentration-dependent manner by $MgCl_2$ added to the medium (Fig. 1A). The matrix $[Mg^{2+}]$ as determined from furaptra fluorescence in a parallel incubation is shown in Fig. 1B. Exogenous Mg^{2+} enters the divalent cation-depleted mitochondria on residual BrA-23187 in exchange for 2 H^+ and requires 150–200 s to reach a steady state (Fig. 1B). Matrix $[Mg^{2+}]$ does not equilibrate with external $[Mg^{2+}]$ under these conditions, since lysis of the mitochondria with detergent results in a large fluorescence signal increase (not shown) (28). With 1 mM $MgCl_2$ added to the medium, matrix $[Mg^{2+}]$ is only ~250 μ M after 200 s (Fig. 1B). Because furaptra measures matrix $[Mg^{2+}]$ directly, the failure of the BrA-23187 to equilibrate $[Mg^{2+}]$ across the inner membrane is not relevant to the present measurements or conclusions.

The swelling shows a lag of 30–40 s, a maximum rate that is maintained for ≥ 50 s, and then a decline as the incubation continues (Fig. 1A) (11). It is known that the K^+/H^+ antiport is strongly affected by pH (6, 11, 37). The rate of swelling in potassium acetate increases without apparent limit as external pH is increased (37). This may be due in part to decreased competition between K^+ and H^+ for an external catalytic site (11), but the K^+/H^+ antiport has also been shown to be inhibited allosterically by matrix protons (6, 11). It is also well established that depletion of matrix divalent cations with A-23187 results in decreased matrix pH (6, 11). However, the mitochondria used in this study were depleted of divalent cations in a KCl medium at pH 7.4 and washed in KCl, conditions that minimize matrix acidification. A parallel incubation with mitochondria loaded with SNARF-1 (Fig. 1C) indicates that when the depleted mitochondria are suspended in potassium acetate medium at pH 7.8, matrix pH is initially 7.55 but increases to pH 7.7 at ~50 s and 7.75 at 100 s (Fig. 1C; see also Fig. 5 in Ref. 11 for BCECF records of a quite similar protocol). This establishes that the ΔpH is –0.10 to –0.05 when the swelling reaches its maximum rate (Fig. 1A) and that the inhibition due to added

Fig. 1. Inhibition of K^+/H^+ antiport activity by matrix free Mg^{2+} concentration ($[\text{Mg}^{2+}]$). Beef heart mitochondria were depleted of divalent cations and loaded with furaptra. Depleted mitochondria were suspended at 0.5 mg/ml in basic reaction medium (pH 7.8) containing indicated concentrations of MgCl_2 . A: K^+/H^+ antiport activity measured by passive swelling in hypotonic K^+ acetate. Light scattering was recorded, and light-scattering variable (β) was calculated. B: matrix $[\text{Mg}^{2+}]$ estimated in a parallel incubation by means of furaptra fluorescence and calculated using a dissociation constant (K_d) of 1.29 mM. C: matrix pH determined under identical conditions with carboxysemaphthorhodofluor-1 (SNARF-1)-loaded mitochondria and calculated as described in Ref. 3. There was no change in pH record in presence of 0.6 mM MgCl_2 .



$[\text{Mg}^{2+}]$ is not complicated by changing H^+ concentration. Also, the K_d for Mg^{2+} -furaptra is relatively insensitive to pH in this range (32).

The fastest rates of swelling (β/min) after the lag phase were determined from Fig. 1A and plotted as the percentage of maximum K^+/H^+ activity vs. the matrix $[\text{Mg}^{2+}]$ at the time these rates were achieved (Fig. 2). The data were fit to a sigmoidal curve, and the concentration that results in 50% inhibition of maximum activity (IC_{50}) of K^+/H^+ activity is 92 μM Mg^{2+} . The average value of matrix $[\text{Mg}^{2+}]$ (670 μM) that is found in our preparations of furaptra-loaded heart mitochondria (29) is indicated in Fig. 2.

Inhibition of K^+/H^+ antiport by $[\text{Ca}^{2+}]$. The procedure used for Mg^{2+} (Fig. 1) was repeated to evaluate inhibition due to Ca^{2+} by adding 0–1.0 mM CaCl_2 to the medium (records not shown). The fastest rates of swelling and the matrix $[\text{Ca}^{2+}]$ corresponding to the time of the fastest swelling rates were determined and plotted as percent K^+/H^+ activity vs. matrix $[\text{Ca}^{2+}]$ (Fig. 3). These data were curve fit to a hyperbola, and a linear Hanes plot indicates an IC_{50} of 2.2 μM for inhibition of K^+/H^+ activity by $[\text{Ca}^{2+}]$.

The experiment in Fig. 4 shows that when Ca^{2+} and Mg^{2+} are present in the medium, the effect of the two cations on K^+/H^+ activity is additive. When added individually, MgCl_2 (0.2 and 0.4 mM) or CaCl_2 (40 μM)

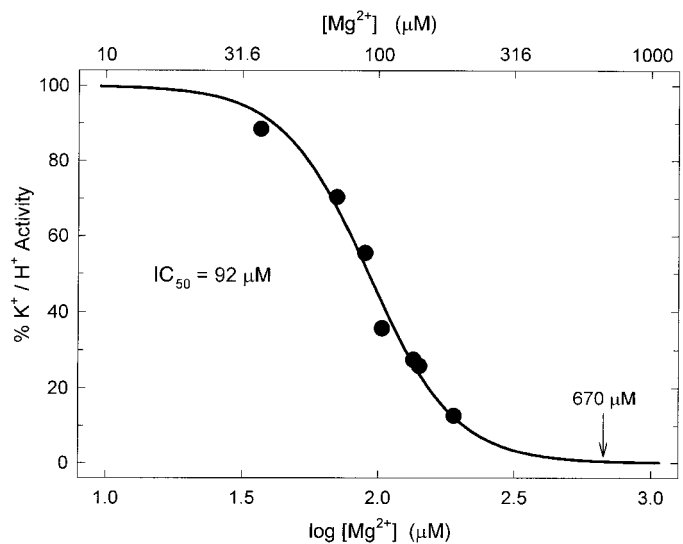


Fig. 2. K^+/H^+ antiport activity as a function of matrix $[\text{Mg}^{2+}]$. Fastest swelling rates after initial lag period were determined at ~ 100 s from traces in Fig. 1A by linear regression analysis. $[\text{Mg}^{2+}]$ values were taken from Fig. 1B at time fastest rate was measured. Data were curve fit to a sigmoidal plot with Graphpad and then normalized, with maximum and minimum set to 100 and 0%, respectively. Concentration that results in 50% inhibition of maximum activity (IC_{50}) of K^+/H^+ activity by matrix $[\text{Mg}^{2+}]$ is 92 μM . Average matrix $[\text{Mg}^{2+}]$ of furaptra-loaded heart mitochondria is 670 μM (arrow) (29).

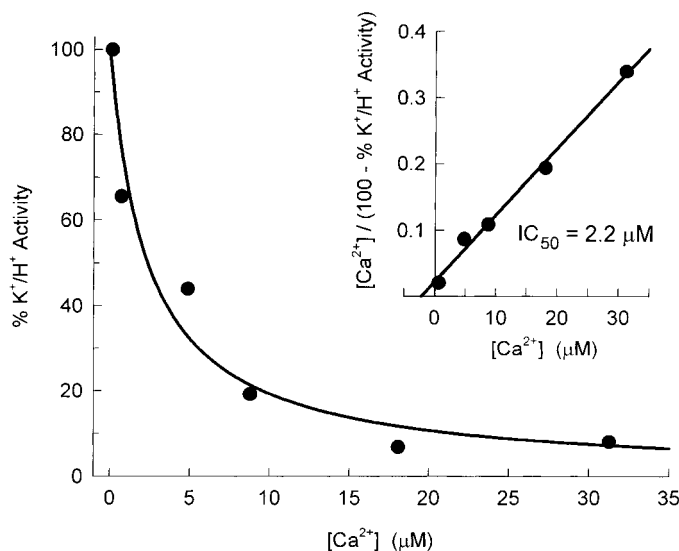


Fig. 3. K^+/H^+ antiport activity as a function of matrix free Ca^{2+} concentration ($[\text{Ca}^{2+}]$). Incubation conditions and calculations were identical to those described in Fig. 1 legend, except CaCl_2 (0–1 mM) replaced MgCl_2 . Fastest swelling rates were determined from light-scattering records, and corresponding matrix $[\text{Ca}^{2+}]$ were calculated from furaptra fluorescence with a K_d of 13.7 μM . Data were curve fit using Graphpad and then normalized, with maximum and minimum set to 100 and 0%, respectively. Inset: Hanes plot. IC_{50} for inhibition of K^+/H^+ activity by matrix $[\text{Ca}^{2+}]$ is estimated to be 2.2 μM .

inhibits the swelling of beef heart mitochondria in a potassium acetate medium significantly. Addition of Ca^{2+} with either concentration of Mg^{2+} results in an even greater inhibition of swelling, and added Mg^{2+} increases the inhibition seen with Ca^{2+} alone (Fig. 4).

Changes in matrix $[\text{Mg}^{2+}]$ with swelling due to P_i accumulation. If matrix $[\text{Mg}^{2+}]$ is to function as a volume sensor, its concentration should change with mitochondrial swelling. Heart mitochondria that have not been depleted of divalent cations show only a minimal change in volume (Fig. 5A), but matrix $[\text{Mg}^{2+}]$ declines from an initial value of ~ 1 mM to ~ 0.7 mM (Fig. 5B) during respiration with succinate and rotenone in a KCl medium. This is apparently due to an increase in Mg^{2+} ligands, such as citrate, and is linked

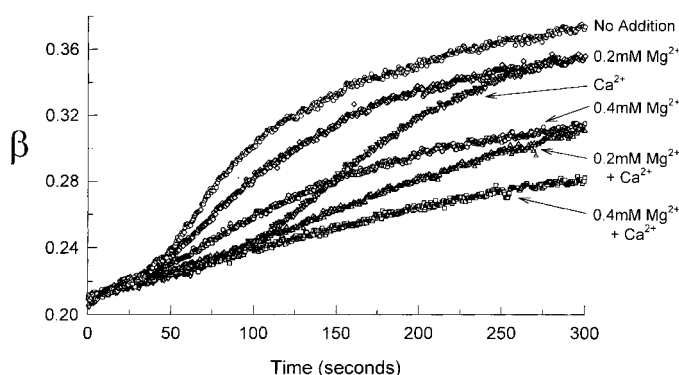


Fig. 4. Mg^{2+} and Ca^{2+} show additive inhibition of K^+/H^+ antiport activity. Conditions and calculations were the same as those described in Fig. 1A with indicated concentrations of MgCl_2 or 40 μM CaCl_2 added to medium.

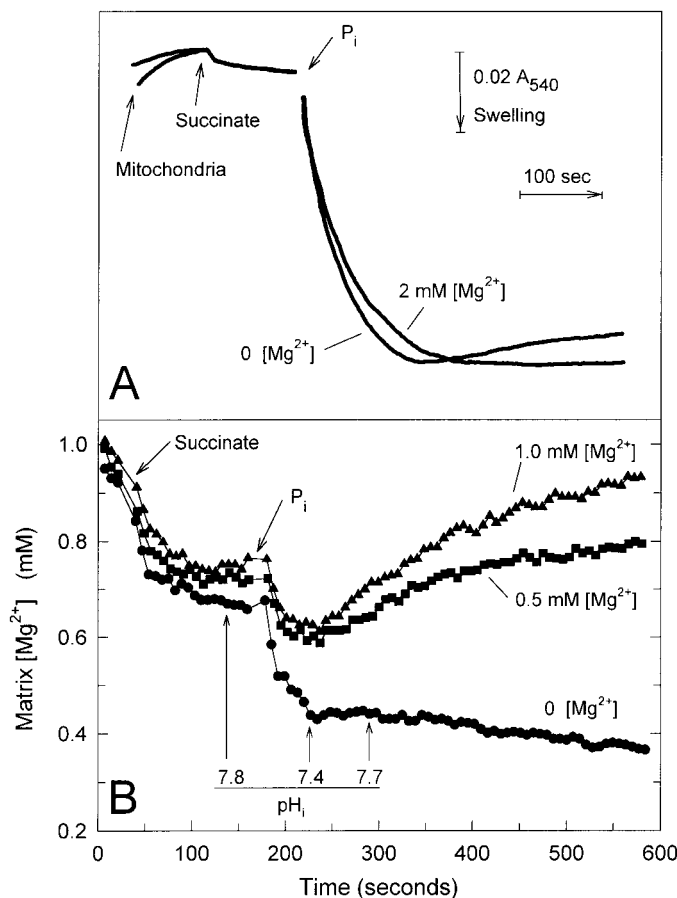


Fig. 5. Effect of external $[\text{Mg}^{2+}]$ on P_i -induced swelling and matrix $[\text{Mg}^{2+}]$. A: furaptra-loaded beef heart mitochondria suspended at 0.5 mg/ml in KCl (0.1 M) containing HEPES (10 mM, pH 7.4), rotenone (3 $\mu\text{g}/\text{ml}$), EGTA (2 mM), and MgCl_2 to buffer $[\text{Mg}^{2+}]$ at 0, 0.5, 1.0, and 2.0 mM after addition of succinate (3.3 mM) and P_i (3.3 mM). Swelling was measured in a DW-2C spectrophotometer and was nearly identical at all 3 external Mg^{2+} concentrations. A_{540} , absorbance at 540 nm. B: matrix $[\text{Mg}^{2+}]$ was estimated in a parallel incubation and calculated from furaptra fluorescence with a K_d of 2.1 mM. Total mitochondrial Mg^{2+} after 600 s was estimated by atomic absorption and showed a decrease of 1 nmol/mg protein for zero external $[\text{Mg}^{2+}]$ and no change for external $[\text{Mg}^{2+}]$ of 0.5 and 1.0 mM. Matrix pH (pH_i) was measured in a parallel incubation with use of BCECF-loaded mitochondria (11).

to a reduction of matrix pyridine nucleotides under these conditions (25, 26).

When no exogenous Mg^{2+} is present, addition of P_i under these conditions causes the mitochondria to accumulate potassium phosphate and swell osmotically (Fig. 5A). As the mitochondria swell, matrix $[\text{Mg}^{2+}]$ decreases to ~ 0.4 mM (Fig. 5B). There is some loss of total Mg^{2+} from the mitochondria under these conditions (8, 25, 29), so the decrease in matrix $[\text{Mg}^{2+}]$ can be due to a reduction in total Mg^{2+} as well as the increased concentration of phosphate. When $[\text{Mg}^{2+}]$ is present in the suspending medium at 0.5–2.0 mM, there is little effect on swelling after P_i addition (Fig. 5A), but matrix $[\text{Mg}^{2+}]$ shows a different pattern of change. Matrix $[\text{Mg}^{2+}]$ decreases initially from 0.7 to ~ 0.6 mM on P_i addition in the presence of 1 mM external Mg^{2+} , but then rebounds and, after 100 s, attains a level compa-

rable to that before the addition of P_i (Fig. 5B). With continued incubation, matrix $[\text{Mg}^{2+}]$ increases to ~ 0.9 mM. A similar response is seen with 0.5 mM external $[\text{Mg}^{2+}]$ (Fig. 5B). These results demonstrate that mitochondria can swell significantly due to the uptake of potassium phosphate without a sustained decrease in matrix $[\text{Mg}^{2+}]$.

The accumulation of phosphate under the conditions of Fig. 5 results from symport of phosphate and H^+ (or P_i/OH^- antiport) on the phosphate transporter. This converts a portion of the ΔpH to $\Delta\psi$ and favors the electrophoretic uptake of K^+ (12) and osmotic swelling. A parallel incubation using mitochondria loaded with BCECF (records not shown) shows that matrix pH increases to 7.8 with succinate respiration, drops rapidly to pH 7.4 on addition of P_i , and rebounds spontaneously to pH 7.7 over the next 60 s (Fig. 5B) as cation uptake restores the ΔpH component.

DISCUSSION

Inhibition of K^+/H^+ antiport by matrix $[\text{Mg}^{2+}]$. The present studies show that the K^+/H^+ antiport activity of divalent cation-depleted heart mitochondria is inhibited by matrix $[\text{Mg}^{2+}]$ with an IC_{50} of 92 μM (Fig. 2). This value is significantly less than $[\text{Mg}^{2+}]$ of 200–400 μM estimated by Garlid (15). The higher value was calculated from an observed IC_{50} for external $[\text{Mg}^{2+}]$ of 50–65 μM in the presence of A-23187 “sufficient to assure equilibrium” with respect to $\text{Mg}^{2+}/2\text{H}^+$ exchange and the assumption that the ΔpH was -0.3 to -0.4 (matrix acid). However, direct measurement with fluorescent probes shows that the ΔpH is dissipated rapidly under these conditions (Fig. 1C). The present IC_{50} values would seem to be the more reliable, because furaptra provides a direct measure of matrix $[\text{Mg}^{2+}]$ that is independent of assumptions regarding ΔpH and does not depend on equilibration of Mg^{2+} by the ionophore. When the measured ΔpH of -0.05 (Fig. 1C) is used in Garlid’s calculation, IC_{50} values of 60–80 μM are obtained. The substantial agreement between these values and our estimate of 92 μM is reassuring in view of our previous reservations regarding the calibration of furaptra fluorescence in terms of absolute matrix $[\text{Mg}^{2+}]$ (28).

Kakar et al. (30) reported an IC_{50} of 300–350 μM for $[\text{Mg}^{2+}]$ inhibition of Rb^+ uptake by a reconstituted K^+/H^+ antiport from rat liver mitochondria. However, this result was obtained using proteolipid vesicles with high leak rates, a very low Rb^+ concentration (1 mM) relative to the Michaelis-Menten constant for transport of this cation (130 mM), and assay conditions that are far from those found in intact mitochondria.

The matrix $[\text{Mg}^{2+}]$ of isolated beef heart mitochondria when loaded with furaptra averages 670 μM and varies with total mitochondrial Mg^{2+} (29). This level of matrix $[\text{Mg}^{2+}]$ would put K^+/H^+ antiport activity at $<1\%$ of maximal on the basis of the plot shown in Fig. 2. For the antiport to respond rapidly to small changes in matrix $[\text{Mg}^{2+}]$, as postulated by the carrier-brake model, $[\text{Mg}^{2+}]$ must be reduced to a level that is on the ascending portion of this activity plot (36). For ex-

ample, a decrease in matrix $[\text{Mg}^{2+}]$ to 200 μM would permit $\sim 10\%$ of maximum antiport activity (Fig. 2).

Is matrix $[\text{Mg}^{2+}]$ an adequate volume sensor? For matrix $[\text{Mg}^{2+}]$ to be the sensor of mitochondrial volume changes, its concentration must change with swelling and contraction. We have shown that matrix $[\text{Mg}^{2+}]$ decreases markedly with hypotonic swelling when heart mitochondria are exposed to a series of KCl buffers of decreasing tonicity (see Fig. 8 in Ref. 25). A K_d of 1.5 mM was used to calculate the matrix $[\text{Mg}^{2+}]$ reported previously (25). We have now recalculated the matrix $[\text{Mg}^{2+}]$ from these experiments using values for the K_d of Mg^{2+} -furaptra that are appropriate for the changing ionic strength of the media and the assumption that matrix K^+ concentration is close to the external concentration when the rapid hypotonic swelling is complete. These calculations show that matrix $[\text{Mg}^{2+}]$ decreases from 1.34 mM in hypertonic (150 mM) KCl to near 100 μM when the mitochondria are extensively swollen (Fig. 6). When the mitochondria are then contracted by addition of KCl to a final concentration of 150 mM, matrix $[\text{Mg}^{2+}]$ increases to within 80% of the original value (Fig. 6).

This indicates that as the matrix increases in volume because of the uptake of water alone, Mg^{2+} does not dissociate from its binding sites in appreciable amounts and matrix $[\text{Mg}^{2+}]$ decreases dramatically. Approximately 97% of the total Mg^{2+} in the matrix is bound (13, 29). It was the extrusion of K^+ after hypotonic swelling that led to the proposal of the Mg^{2+} carrier-brake model (14), and the data of Fig. 6 are in line with this model. However, hypotonic swelling would not be expected to occur in mitochondria in situ.

In contrast to hypotonic swelling, osmotic swelling caused by the respiration-dependent accumulation of K^+ and acetate (which is not an effective Mg^{2+} ligand) does not result in decreased matrix $[\text{Mg}^{2+}]$ (29). Furaptra fluorescence shows that matrix $[\text{Mg}^{2+}]$ increases slightly from its original value of 0.57 mM to 0.65 mM

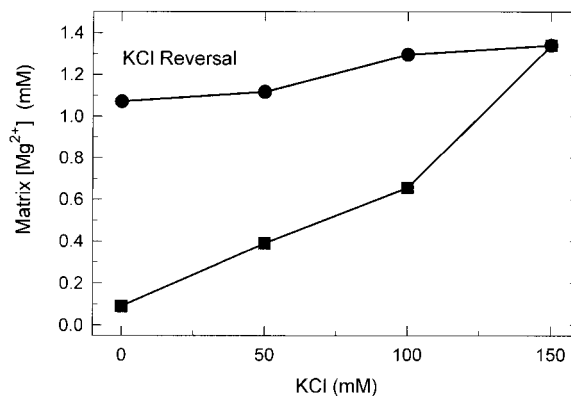


Fig. 6. Effects of hypotonic swelling on matrix $[\text{Mg}^{2+}]$. Values for matrix $[\text{Mg}^{2+}]$ reported in Fig. 8 of Ref. 25 were recalculated using K_d of 0.17, 1.2, 2.1, and 3.35 mM for Mg^{2+} -furaptra in presence of 0, 50, 100, and 150 mM KCl, respectively (28). Furaptra-loaded mitochondria were suspended at 0.5 mg/ml in HEPES buffer (10 mM, pH 7.4) containing rotenone (3 $\mu\text{g}/\text{ml}$), EGTA (2 mM), and indicated concentration of KCl. Change in matrix $[\text{Mg}^{2+}]$ when mitochondria are contracted by addition of KCl to a final concentration of 150 mM is also shown.

during extensive swelling (change in absorbance at 540 nm of 0.14) in acetate (see Fig. 6 in Ref. 29). This suggests that K^+ can displace enough bound Mg^{2+} under these conditions to keep matrix $[\text{Mg}^{2+}]$ effectively buffered. The lack of change in matrix $[\text{Mg}^{2+}]$ with acetate swelling is clearly not consistent with its proposed role as the mitochondrial volume sensor (14, 15).

Garlid (14) showed that anions with a high affinity for Mg^{2+} (mainly citrate and phosphate) promote the loss of matrix K^+ from liver mitochondria, presumably by activation of K^+/H^+ antiport. The accumulation of citrate has been shown to decrease matrix $[\text{Mg}^{2+}]$ in liver mitochondria (13). However, heart mitochondria lack the dicarboxylate (31) and the tricarboxylate (2) transporters. This prevents phosphate from exiting the matrix of heart mitochondria in exchange for substrate anions, as it does in liver, and makes significant uptake of citrate as a K^+ counterion during osmotic swelling unlikely. It follows that phosphate would be the most likely anion that could decrease matrix $[\text{Mg}^{2+}]$ by its accumulation in heart mitochondria in situ.

Recent studies have confirmed that increasing matrix phosphate decreases the proportion of total Mg^{2+} that is free in the matrix (29). When heart mitochondria swell as a result of phosphate uptake (Fig. 5), matrix $[\text{Mg}^{2+}]$ drops to $\sim 400 \mu\text{M}$ if external Mg^{2+} is absent. However, a change of this magnitude would increase K^+/H^+ antiport activity from $<1\%$ to only $\sim 2\%$ of maximum (Fig. 2). In addition, when physiological levels of external Mg^{2+} are present, the decrease in matrix $[\text{Mg}^{2+}]$ is nowhere near this large, and even though the matrix has been expanded to a considerable extent, there is no sustained decrease in matrix $[\text{Mg}^{2+}]$ (Fig. 5B).

Cellular ATP is normally nearly saturated with Mg^{2+} (13), but under pathological conditions, ATP is hydrolyzed and $[\text{Mg}^{2+}]$ increases. When heart cells are subjected to hypoxia and depleted of ATP, cytosolic $[\text{Mg}^{2+}]$ more than doubles, increasing from 1.3 to 2.8 mM (44). Under pathological conditions where $[\text{Ca}^{2+}]$ and $[\text{Mg}^{2+}]$ may be elevated, inhibition, rather than activation, of K^+/H^+ antiport activity would be expected. It has also been suggested that $[\text{Mg}^{2+}]$ can serve as a volume transducer in intact cells, but changes in $[\text{Mg}^{2+}]$ with cell volume have not been demonstrated (41).

These considerations make it seem unlikely that a decrease in matrix $[\text{Mg}^{2+}]$ that would permit significant activation of the K^+/H^+ antiport can be attained in vivo by accumulation of Mg^{2+} ligands, such as P_i . Because matrix $[\text{Mg}^{2+}]$ must fluctuate more dramatically than these studies indicate to provide precise regulation of K^+/H^+ antiport activity, the concept that it can function as the sensor for volume homeostasis must be called into question. Activation of the electrophoretic anion channel would also seem unlikely, because the IC_{50} for its inhibition by $[\text{Mg}^{2+}]$ has been reported to be $38 \mu\text{M}$ at pH 7.4 and $90 \mu\text{M}$ at pH 7.8 (4).

An additional argument against a role for matrix $[\text{Mg}^{2+}]$ as the volume sensor is that it can vary with metabolic conditions in the absence of changes in

matrix volume (25). For example, isolated heart mitochondria respiring with glutamate and malate show a reversible increase in matrix $[\text{Mg}^{2+}]$ of 0.1–0.2 mM while undergoing a state 4–3–4 cycle (see Figs. 4 and 5 in Ref. 25). These changes are accompanied by changes in the redox state of mitochondrial pyridine nucleotides and appear to depend on alterations in matrix citrate content. The decrease in matrix $[\text{Mg}^{2+}]$ shown in Fig. 5 before phosphate addition is also not dependent on swelling.

Inhibition of K^+/H^+ antiport by matrix $[\text{Ca}^{2+}]$. K^+/H^+ antiport activity is also inhibited by $[\text{Ca}^{2+}]$ with an IC_{50} of $2.2 \mu\text{M}$ (Fig. 3). Garlid (15) put this value at 50–90 μM (calculated from observed values of 12–18 μM), but our lower estimate seems more appropriate for the reasons discussed above for $[\text{Mg}^{2+}]$. Mitochondrial K^+/H^+ antiport was reported to be inhibited by Ca^{2+} in heart mitochondria treated with A-23187 (43), but Nakashima et al. (36) concluded that Ca^{2+} has no effect on this activity unless Mg^{2+} is first depleted. It has also been reported that Ca^{2+} in the presence of Mg^{2+} stimulates K^+/H^+ antiport activity indirectly in rat liver mitochondria (36). The present studies (Fig. 4) indicate that Ca^{2+} can inhibit K^+/H^+ antiport activity in the presence of Mg^{2+} and Mg^{2+} can increase the inhibition with Ca^{2+} .

It has been shown that Ca^{2+} can increase net uptake of K^+ and swelling of mitochondria, and these Ca^{2+} -dependent volume increases have been related to increased respiration rate and hormone signaling to the mitochondria (see Ref. 21 for review). Matrix $[\text{Ca}^{2+}]$ has been reported to increase in mitochondria in situ and can reach levels of $\geq 4 \mu\text{M}$ (40). It would therefore not seem unreasonable for this cation to have significant effects on K^+/H^+ antiport activity and take part in mitochondrial volume homeostasis. However, it is not clear why changes in $[\text{Ca}^{2+}]$ would affect the antiport if it is already fully inhibited by matrix $[\text{Mg}^{2+}]$ under normal metabolic conditions.

Alternative models for mitochondrial volume regulation. A volume control system consists of a sensor that detects volume changes, a transducer that will convey a signal to a volume-regulatory transporter, and a change in the activity of the transporter (22, 23). Because our study indicates that matrix $[\text{Mg}^{2+}]$ is unlikely to perform the role of volume sensor and transducer, other mechanisms for mitochondrial volume sensing and control must be considered.

Volume-sensing elements may act by concentration or mechanical mechanisms (22, 23). A change in the matrix protein concentration (relief of “macromolecular crowding”) by itself may act as a signal by affecting transducing elements, such as a kinase or phosphatase, that would regulate a transporter (34). An obvious mechanical signal could involve contact between the inner and outer membranes functioning to activate a volume-regulatory transporter. Alternatively, components analogous to cellular stretch receptors may be present to sense swelling of the mitochondria (7). The fact that *N,N'*-dicyclohexylcarbodiimide inhibition of K^+/H^+ antiport activity in Mg^{2+} -depleted mitochondria

occurs in hypotonic, but not in isotonic, media (17) suggests that stretching of the membrane or dilution or dissociation of screening matrix proteins may be involved in exposing the reactive sites.

Whereas there is considerable evidence that a K^+/H^+ antiport functions as a volume control transporter (15), the possibility that other mechanisms may be involved should be considered. Mitochondrial K^+/H^+ antiport activity requires extensive depletion of matrix $[Mg^{2+}]$, elevated pH, and hypotonic conditions for its demonstration, and there have always been questions as to whether it might be an artifact derived from another transporter by the conditions imposed (see the discussion in Ref. 9). When Mitchell (35) proposed that electroneutral exchangers must exist to allow efflux of cations against the membrane potential, he also suggested an alternative mechanism, namely, that an occasional bursting and resealing of the inner mitochondrial membrane would allow equilibration of matrix solutes with those in the cytosol. It seems possible that matrix K^+ and Mg^{2+} might be equilibrated by diffusion through a transient pore, such as the mitochondrial permeability transition pore (20), especially since ion-conducting substates of the pore appear to be available. If the permeability transition pore or a substate indeed opens and closes, or flickers, under normal metabolic conditions, as has been suggested by several studies (1, 24, 38), it is possible that it could be triggered at appropriate times in each individual mitochondrion to allow equilibration of small solutes. Gunter and Pfeiffer (20) considered this mechanism and found that it was not unreasonable from an energetic standpoint.

It is obvious that further work is necessary to clarify exactly the mechanism of mitochondrial volume regulation and the physiological role of the K^+/H^+ antiport.

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